

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10	U04						***** GATE VALVES *****			B1	ALL BOLTS ARE METRIC DIA.
20										FW3	VIKING JOHNSON COUPLINGS SHALL ONLY BE USED ON 'WF' FIREWATER SERVICE
20	U04		GATE	GA6B1	0.5	1.5	GATE 150# SCRD FNPT BS1400 LG2 BODY AND TRIM HW	0	0		
30	U04		GATE	G01AC41	2	8	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW	0	0	GA3	SPIRAL WOUND GASKETS SHALL ONLY BE USED BETWEEN CARBON STEEL FLANGES AND CARBON STEEL VALVES.
40	U04	VA5	GATE	G01AC96	2	16	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW 2800MM EXT	0	0		
50	U04	VA5	GATE	G01AC99	2	16	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW 3300MM EXT	0	0	ND9	USE ONLY WITH CONSENT OF MATERIALS ENGINEER.
60	U04	VA5	GATE	G01AC95	2	16	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW 3700MM EXT	0	0	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
70											
70	U04						***** GLOBE VALVES *****				
80											
80	U04		GLOBE	L01AC41	2	6	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y HW	0	0		
90	U04		GLOBE	L01AC42	8	8	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y GO	0	0		
100											
100	U04						***** BUTTERFLY VALVES *****				
110											
110	U04		BUTTERFLY	T01AC61	10	24	BUTTERFLY 150# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO	0	0		
120											
120	U04						***** PIPE *****				
130											
130	U04	PA0	PIPE	5713100	2	24	PIPE SDR 9 HD POLYETHYLENE PE 100 WATER HOSTALEN CRP 100 BLACK RESIST CR ISO 4427	0	0		
140											
140	U04	PA0					***** FITTINGS *****				
150											
150	U04		ELBOW	5713446	2	10	BEND 90 DEG SEGMENTED & MACHINED TYPE SDR 9 HD POLYETHYLENE PE 100 WATER HOSTALEN CRP100 BLACK RESIST CR ISO 4427	0	0		
160	U04		ELBOW	5713442	12	24	BEND 90 DEG SEAMLESS TYPE SDR 9 HD POLYETHYLENE PE 100 WATER HOSTALEN CRP100 BLACK RESIST CR ISO 4427	0	0		
170	U04		ELBOW	5713445	2	10	BEND 45 DEG SEGMENTED & MACHINED TYPE SDR 9 HD POLYETHYLENE PE 100 WATER HOSTALEN CRP100 BLACK RESIST CR ISO 4427	0	0		
180	U04		ELBOW	5713441	12	24	BEND 45 DEG SEAMLESS TYPE SDR 9 HD POLYETHYLENE PE 100 WATER HOSTALEN CRP100 BLACK RESIST CR ISO 4427	0	0		
190	U04		INDUC BEND	5713439	2	2	BEND 22.5 DEG SEGMENTED & MACHINED TYPE SDR 9 HD POLYETHYLENE PE 100 WATER HOSTALEN CRP100 BLACK RESIST CR ISO 4427	0	0		
200											
200	U04		TEE	5713449	2	24	TEE EQUAL MACHINED TYPE SDR 9 HD POLYETHYLENE PE 100 WATER HOSTALEN CRP100 BLACK RESIST CR ISO 4427	0	0		
210	U04		TEE RED	5713450	2	18	TEE REDUCING MACHINED TYPE SDR 9 HD POLYETHYLENE PE 100 WATER HOSTALEN CRP100 BLACK RESIST CR ISO 4427	3	24		
220											
220	U04		CAP	5713443	2	24	CAP MACHINED TYPE SDR 9 HD POLYETHYLENE PE 100 WATER HOSTALEN CRP 100 BLACK RESIST CR ISO 4427	0	0		
230											
230	U04		RED CONC	5713447	2	14	REDUCER CONC MACHINED TYPE SDR 9 HD POLYETHYLENE PE 100 WATER HOSTALEN CRP100 BLACK RESIST CR ISO 4427	3	18		
240	U04		RED CONC	5713447	16	16	REDUCER CONC MACHINED TYPE SDR 9 HD POLYETHYLENE PE 100 WATER HOSTALEN CRP100 BLACK RESIST CR ISO 4427	24	24		
250	U04		RED ECC	5713448	2	10	REDUCER ECC ELONGATED TYPE SDR 9 HD POLYETHYLENE PE 100 WATER HOSTALEN CRP100 BLACK RESIST CR ISO 4427	3	12		
260											
260	U04		LATERAL	5713689	4	10	LATERAL 45 DEG SDR 9 HD POLYETHYLENE PE 100 WATER HOSTALEN CRP100 BLACK RESIST CR ISO 4427	0	0		
270											
270	U04		Coupling	5713444	2	24	COUPLING ELECTRO FUSION SDR 9 HD POLYETHYLENE PE 100 WATER HOSTALEN CRP100 BLACK RESIST CR ISO 4427	0	0		
280											
280	U04	FW3	Coupling	5719244	3	24	MECHANICAL COUPLING VIKING JOHNSON DISMANTLING JOINT (FOR SDR 9)TYPE 150#FF	0	0		
290	U04	ND9	Coupling	5713177	24	24	MECHANICAL COUPLING TEEKAY DISMANTLING JOINT,TYPE VI EPOXY COATED STEEL, 600MM AXIFLEX TYPE IV SEALING GASKET, 150#RF	0	0		
300											
300	U04						***** FLANGES *****				
310											
310	U04		FLANGE	5713451	2	24	HP FLANGE CONNECTION FOR FULL PRESSURE LOAD SDR 9 HD POLYETHYLENE C/W BACKING RING PE 100 WATER HOSTALEN CRP100 ISO 4427	0	0		
320											
320	U04						***** PUDDLE FLANGE SPOOL ADAPTORS *****				
330											
330	U04		FLG ADAPTER	5713492	2	2	SPOOL - PUDDLE (ANCHOR) FLANGE WITH STUB FLANGE C/W BACKING RING AND 2" PIPE SDR 9 HD PE 100 HOSTALEN CRP100 ISO 4427	0	0		
340	U04		FLG ADAPTER	5713491	3	3	SPOOL - PUDDLE (ANCHOR) FLANGE WITH STUB FLANGE C/W BACKING RING AND 3" PIPE SDR 9 HD PE 100 HOSTALEN CRP100 ISO 4427	0	0		
350	U04		FLG ADAPTER	5713481	4	4	SPOOL - PUDDLE (ANCHOR) FLANGE WITH STUB FLANGE C/W BACKING RING AND 4" PIPE SDR 9 HD PE 100 HOSTALEN CRP100 ISO 4427	0	0		

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT FUSION SEAL WELDING - NONE PIPE BENDS - NONE THREAD SEALANT - NONE ELECTRIC SOCKET WELDING OR BUTT WELDING AS APPLICABLE	CORR. ALLOW. 0 MM	DESIGN: DESIGN PRESSURE RANGE AS PER PIM-SU-5112-TCO CODE - SHALL BE ADVISED BY TCO SME FLUID CODES - PIM-SU-5112-TCO	MATERIAL: HDPE PE100 SDR 7.4 ID COLOR: BLUE FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 40 °C	12/16/2024	BY OZAC	U04	Page 1 of 2	150PE4

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360	U04		FLG ADAPTER	5713482	6	6	SPOOL - PUDDLE (ANCHOR) FLANGE WITH STUB FLANGE C/W BACKING RING AND 6" PIPE SDR 9 HD PE 100 HOSTALEN CRP100 ISO 4427	0	0	
370	U04		FLG ADAPTER	5713483	8	8	SPOOL - PUDDLE (ANCHOR) FLANGE WITH STUB FLANGE C/W BACKING RING AND 8" PIPE SDR 9 HD PE 100 HOSTALEN CRP100 ISO 4427	0	0	
380	U04		FLG ADAPTER	5713484	10	10	SPOOL - PUDDLE (ANCHOR) FLANGE WITH STUB FLANGE C/W BACKING RING AND 10" PIPE SDR 9 HD PE 100 HOSTALEN CRP100 ISO 4427	0	0	
390	U04		FLG ADAPTER	5713485	12	12	SPOOL - PUDDLE (ANCHOR) FLANGE WITH STUB FLANGE C/W BACKING RING AND 12" PIPE SDR 9 HD PE 100 HOSTALEN CRP100 ISO 4427	0	0	
400	U04		FLG ADAPTER	5713486	14	14	SPOOL - PUDDLE (ANCHOR) FLANGE WITH STUB FLANGE C/W BACKING RING AND 14" PIPE SDR 9 HD PE 100 HOSTALEN CRP100 ISO 4427	0	0	
410	U04		FLG ADAPTER	5713487	16	16	SPOOL - PUDDLE (ANCHOR) FLANGE WITH STUB FLANGE C/W BACKING RING AND 16" PIPE SDR 9 HD PE 100 HOSTALEN CRP100 ISO 4427	0	0	
420	U04		FLG ADAPTER	5713488	18	18	SPOOL - PUDDLE (ANCHOR) FLANGE WITH STUB FLANGE C/W BACKING RING AND 18" PIPE SDR 9 HD PE 100 HOSTALEN CRP100 ISO 4427	0	0	
430	U04		FLG ADAPTER	5713489	20	20	SPOOL - PUDDLE (ANCHOR) FLANGE WITH STUB FLANGE C/W BACKING RING AND 20" PIPE SDR 9 HD PE 100 HOSTALEN CRP100 ISO 4427	0	0	
440	U04		FLG ADAPTER	5713490	24	24	SPOOL - PUDDLE (ANCHOR) FLANGE WITH STUB FLANGE C/W BACKING RING AND 24" PIPE SDR 9 HD PE 100 HOSTALEN CRP100 ISO 4427	0	0	
450										
450	U04						***** PUDDLE FLANGE SPOOLS *****			
460										
460	U04		PUDDLE FLG	5597715	4	24	PUDDLE (ANCHOR) FLANGE WITH PIPE SDR 9 HD PE 100 HOSTALEN CRP100 ISO 4427 LENGTH 750MM	0	0	
470										
470	U04						***** GASKETS *****			
480										
480	U04		GASKET	5664201	2	24	GASKET 150# RF FLAT RING PTFE	0	0	
490	U04	GA3	GASKET	5673043	2	24	GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
500										
500	U04	B1					***** BOLTING *****			
510										
510	U04		BOLTING	5675768	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) W/ 2 NUTS + LTCS WASHERS	0	0	
520										
520	U04	ND9		5675768-NUT WASHER	0.5	99	NUT WASHER (2 EACH OFF) FOR 5675768 STUD-BOLT	0	0	
530										
530	U04		SPEC BLIND	5749582	2	14	SPECTACLE BLIND 150# RF LTCS A 516 GR.70 GALV A153	0	0	
540										
540	U04		PADDLE	5750195	16	24	PADDLE SPACER 150# RF LTCS A516 GR.70 GALV A153	0	0	
550										

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		ALL SIZES - BUTT FUSION SEAL WELDING - NONE PIPE BENDS - NONE THREAD SEALANT - NONE ELECTRIC SOCKET WELDING OR BUTT WELDING AS APPLICABLE	CORR. ALLOW. 0 MM	DESIGN: DESIGN PRESSURE RANGE AS PER PIM-SU-5112-TCO CODE - SHALL BE ADVISED BY TCO SME FLUID CODES - PIM-SU-5112-TCO	MATERIAL: HDPE PE100 SDR 7.4 ID COLOR: BLUE FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 40 °C	12/16/2024	BY OZAC	U04	Page 2 of 2	150PE4